Memory Forensics FTK Imager + Volatility

Memory forensics is the practice of capturing the active memory (RAM) of a device and subsequently examining the captured data for signs of malicious software or unauthorized activities. In contrast to traditional hard-disk forensics, which involves cloning the entire file system of a device and recovering all files for analysis, memory forensics concentrates on the specific programs that were actively running on the device at the time the memory snapshot was taken.

Random Access Memory (RAM) is a critical component of a device, and its size is often specified in the technical details of a computer or laptop, alongside the hard disk capacity. For instance, you might see specifications like "2TB hard disk" and "32GB RAM" when purchasing a new laptop.

When you open an application on a device, a portion of the available RAM is allocated to that application to ensure its smooth operation. If you open multiple applications simultaneously, the device must allocate more RAM to accommodate them. This is one reason why devices with larger RAM capacities tend to perform better, as they can efficiently manage and run multiple applications concurrently, resulting in enhanced speed and responsiveness.

Volatile data, in the context of incident response and digital forensics, is information that resides in the volatile memory (RAM) of a computing device. It's a critical consideration for Digital Forensic responders because when dealing with a compromised device, one of the initial reactions might be to power off the device in an attempt to contain the security threat.

However, this approach poses a significant challenge. When a device is powered off, any malware or malicious software that has been actively running on the device is stored in the device's memory. All network connections, active processes, and data residing in memory are lost. This is due to the temporary nature of volatile data, which is not written to a persistent (non-volatile) storage medium (such as a hard drive) and is constantly changing in memory. As a result, shutting down the device leads to the loss of valuable evidence that could aid in investigating the compromise.

To effectively respond to an incident without destroying **vital volatile data**, it is advisable to opt for network containment. Network containment involves isolating the compromised

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device from the rest of the network. By doing so, the incident is securely contained, and valuable evidence is preserved within the device's volatile memory, which can then be analysed to understand the nature of the compromise and identify potential threats. This approach ensures that the volatile data in memory remains intact and accessible for forensic examination, ultimately aiding in the investigation and resolution of security incidents.

What is a Memory Dump?

A memory dump, also known as a RAM dump, is a snapshot of a computing device's memory (Random Access Memory) at a specific point in time. This snapshot is captured for the purpose of memory analysis. When a memory dump is obtained, it contains data related to all the running processes and the contents of the device's RAM at the moment the capture was taken. Memory dumps are valuable in digital forensics and incident response because they allow investigators to examine the state of a device's memory and the data it holds at the time of capture. This analysis can provide insights into the activities, processes, and potentially malicious software that were present on the device during the capture, aiding in investigations and security incident responses.

FTK Imager that we used for our AlienImage lab, is a tool essential in the field of digital forensics. It serves the crucial role of creating exact duplicates of original evidence without introducing any alterations. This non-intrusive approach ensures that the integrity of the original evidence remains intact, enabling investigators to perform forensic data acquisition efficiently.

One of the remarkable advantages of FTK Imager is its ability to copy data quickly. This rapid data duplication process facilitates the preservation of digital evidence, ensuring that it remains unchanged throughout the acquisition phase. This untouched state is pivotal for maintaining the evidentiary value of the data, as even minor modifications could compromise its authenticity.

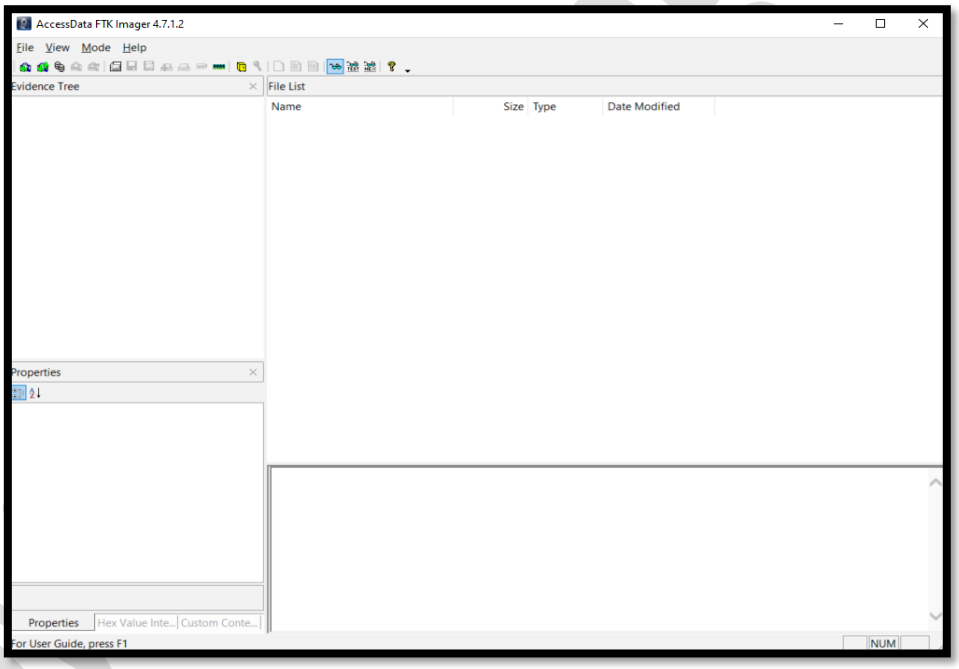
In addition to its primary function of creating forensic images, FTK Imager offers an inbuilt integrity checking feature. This functionality plays a critical role in ensuring the reliability of the acquired data. When an image of the original evidence is generated, the software

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simultaneously computes a cryptographic hash value. This hash value is akin to a digital fingerprint, representing the unique content of the evidence.

The integrity checking function generates a hash report that includes these hash values, allowing investigators to verify the data's integrity. By comparing the hash values before and after the creation of the forensic image, forensic analysts can confirm that the acquired data remains unchanged. Any discrepancies between these hash values would signal potential tampering or corruption of the evidence. This feature provides an additional layer of confidence in the integrity and authenticity of the acquired data, a fundamental aspect of any digital forensic investigation.

FTK IMAGER



Memory capture, a pivotal step in digital forensics, involves preserving the contents of volatile memory in a non-volatile storage medium for subsequent examination. This process ensures that valuable data residing in a computer's volatile memory, which would otherwise be lost after a system reboot, is securely retained for investigative purposes.

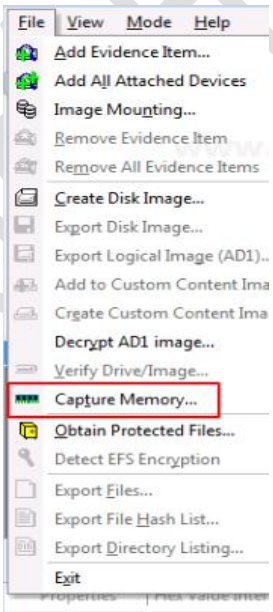
The success of a memory analysis hinges on the accurate and non-corruptive acquisition of volatile memory data. To accomplish this, forensic investigators must exercise caution in their

decisions when collecting this ever-changing data. Any missteps during the acquisition process could result in the loss or corruption of the vital volatile memory image.

Now, let's delve into the process of capturing memory. To initiate memory capture, follow these steps:

1. Open FTK Imager (Make sure you are using the **IMAGER** tool).
2. Navigate to the "File" menu.
3. Select the "Capture Memory" option.

This procedure will commence the memory capture process, allowing forensic professionals to secure and retain volatile memory data effectively. This acquired memory image can then be examined and analysed as part of a broader digital forensic investigation, shedding light on critical information that would otherwise be lost with a system reboot.

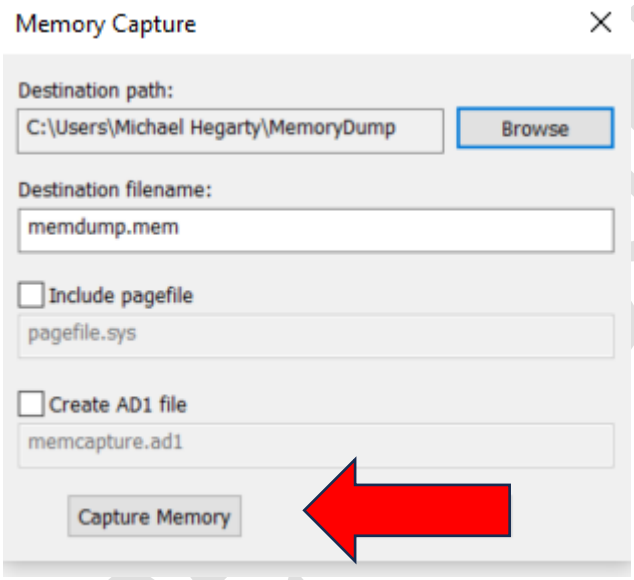


Next, the forensic investigator needs to specify the destination path and provide a suitable name for the destination file where the captured memory will be stored. This critical step ensures that the acquired memory image is saved in an organized and secure location, ready for in-depth analysis.

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After configuring the destination path and file name, click on the "Capture Memory" option to initiate the memory capture process. This action will begin the transfer of volatile memory data to the designated storage location, preserving it for subsequent analysis and investigation.

Properly managing the destination path and file name is essential, as it allows forensic professionals to maintain a systematic and organized approach to digital forensics, ensuring the accuracy and integrity of the collected evidence.

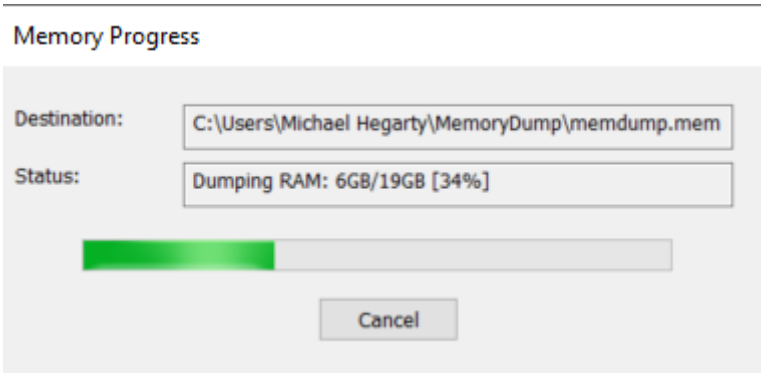


Exercise patience and allow some time for the volatile memory to be captured thoroughly. The duration of this process may vary depending on factors such as the amount of RAM, system performance, and the forensic software being used.

Waiting for a few minutes ensures that the software can effectively acquire the volatile memory without any disruptions or errors. The investigator should refrain from any actions that may interfere with the capture process during this time.

This patience and attention to detail are vital to maintain the integrity and accuracy of the captured memory, which is essential for the subsequent forensic analysis and investigation.

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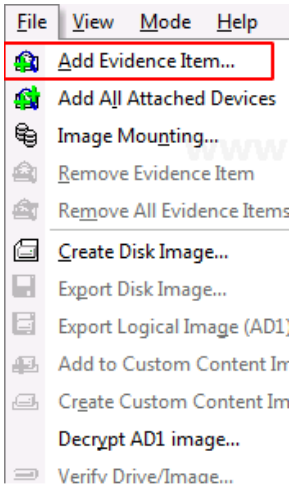


Analysing Image Dump

To begin the analysis of the captured RAW image, you will need to add the evidence item to your forensic software. Follow these steps to do so:

1. Open FTK Imager.
2. Click on "File" in the top menu.
3. Select "Add Evidence Item."

This action will prompt you to browse and locate the RAW image file you previously captured. Once you've added the evidence item, you can proceed with the analysis of the memory dump.

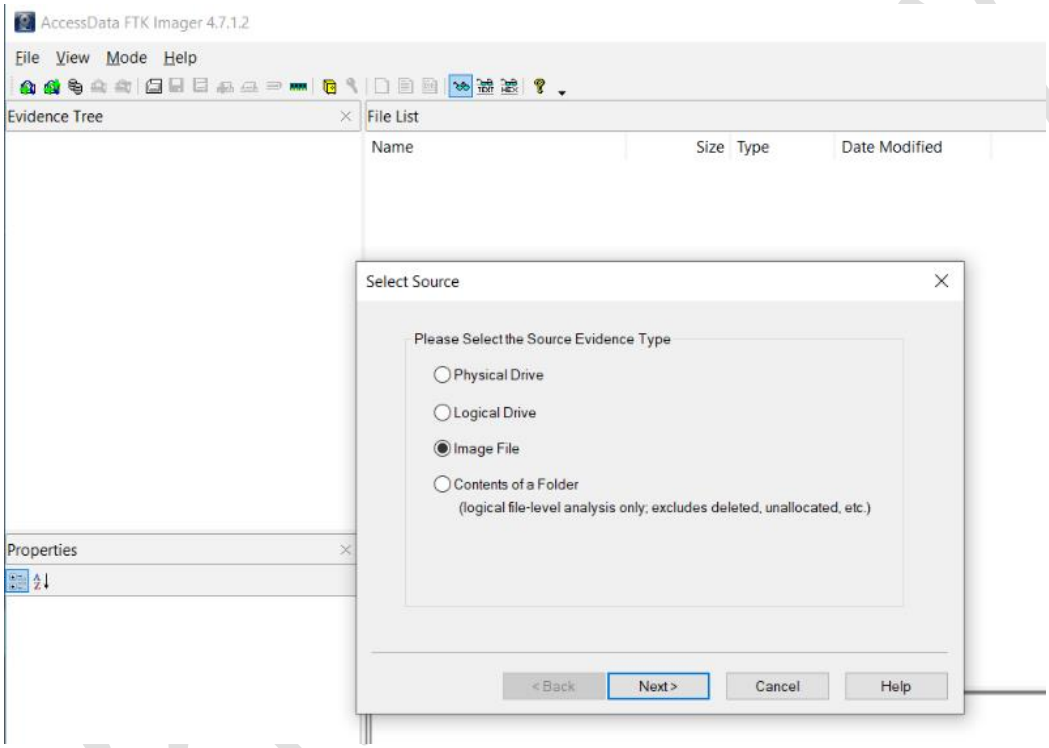


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When you have added the evidence item and need to select the source of the dump file you have created, follow these steps:

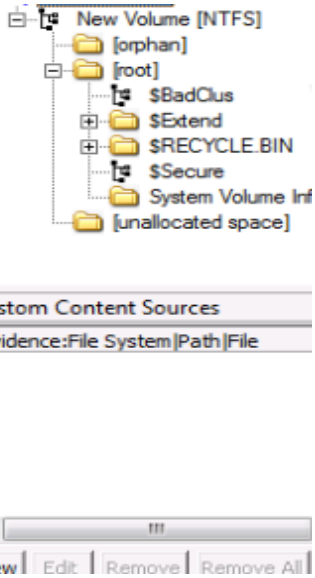
1. In FTK Imager, click on "File" in the top menu.
2. Choose "Add Evidence Item."
3. In the "Add Evidence Item" window, select the "Image File" option.
4. Click "Next" to continue. This will allow you to specify the source file for analysis.



Choose the path of the image dump that you have captured by clicking on Browse.

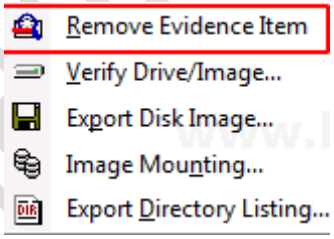
After attaching the image dump to the analysis section in FTK Imager, you will see an evidence tree that contains the contents of the files within the image dump. This may include deleted and overwritten data, which can be crucial for forensic analysis.

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This is an example, yours will be different.

To analyse other things further, we will now remove this evidence item by right-clicking on the case and click on **Remove Evidence Item.**



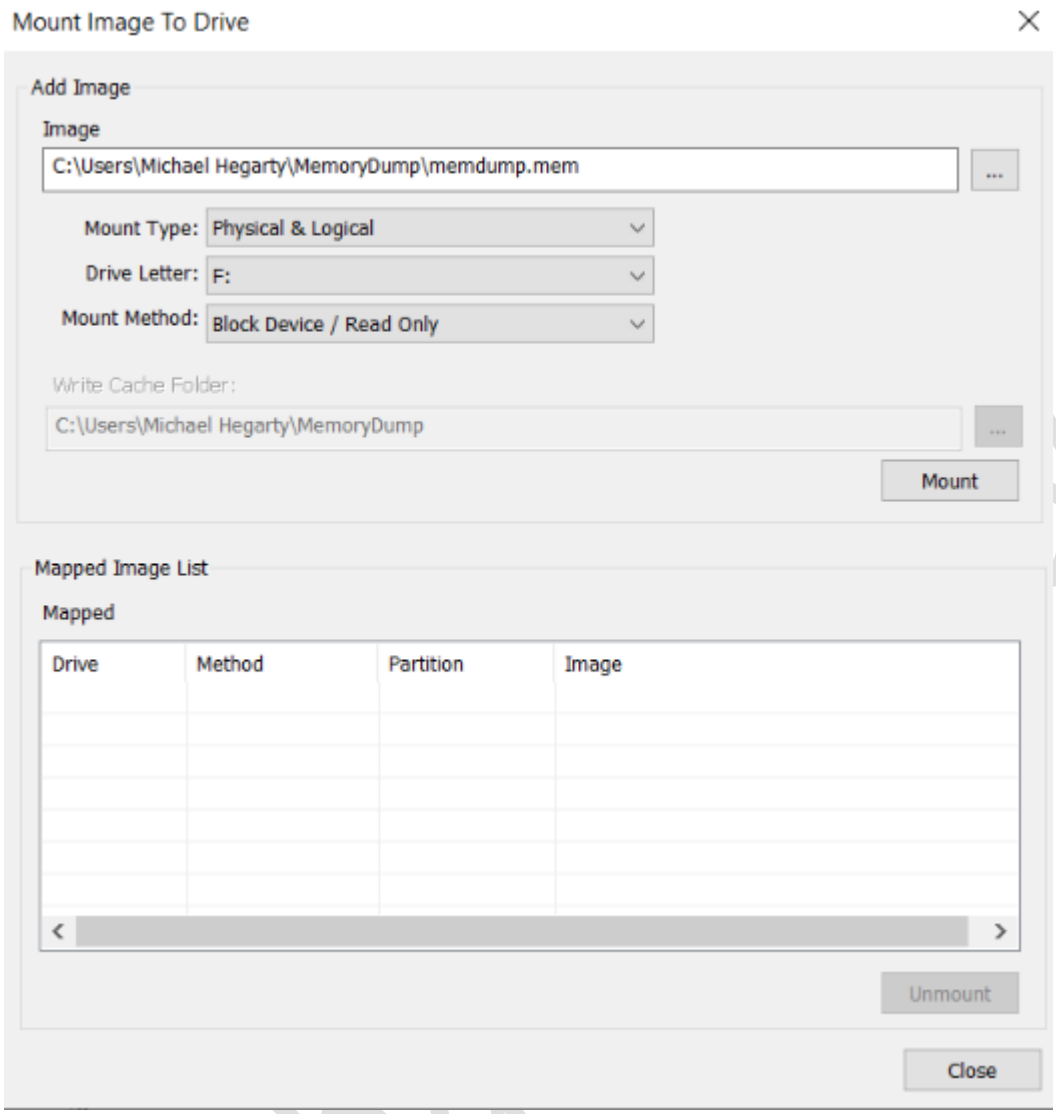
Mounting Image to Drive

To mount the image as a drive in your system, follow these steps:

1. Click on "File" in the FTK Imager menu.
2. Select "Image Mounting" from the dropdown menu.

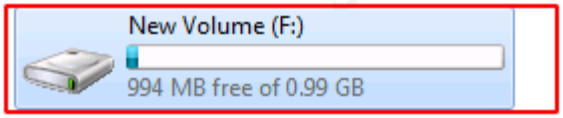
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After clicking "Image Mounting" in the FTK Imager menu, follow these steps:

1. In the "Mount Image to Drive" window, click the "Add" button.
2. Locate the image file you want to mount on your computer and select it.
3. Once you've added the image file, click the "Mount" button to proceed.



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Another excellent tool for Memory Forensics is https://www.volatilityfoundation.org/

Volatility Workbench is a user-friendly graphical interface built upon the well-known Volatility tool, designed for analysing memory dump artifacts. It is open-source, freely available, and designed to operate on Windows-based operating systems. Volatility Workbench seamlessly interacts with .CFG configuration files, which contain essential metadata pertaining to the memory dump file. Starting from version 2, it has extended its support to include memory dumps from Mac and Linux systems, allowing users to select these options from the Profiles folder.

There is a version contained in the folder you downloaded, or it can be also found @ https://www.cyberpunk.rs/volatility-workbench-gui-for-volatility-framework

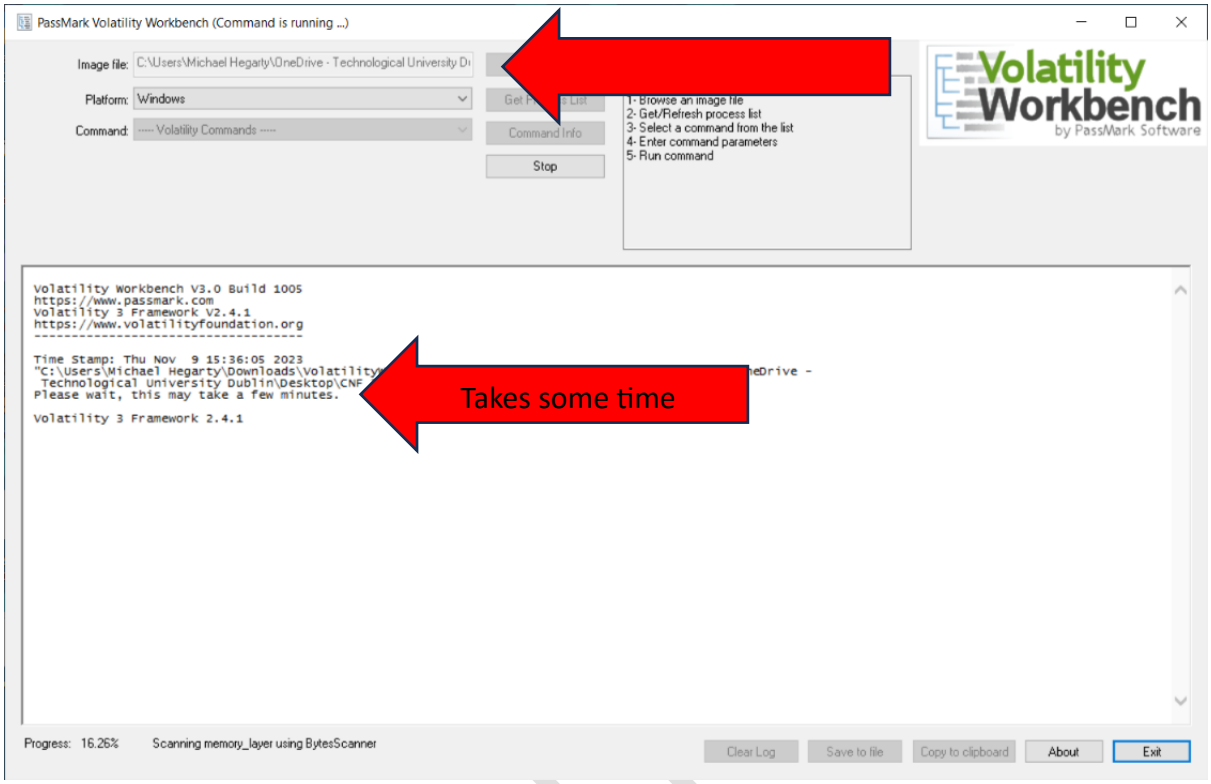
Notable features of Volatility Workbench include:

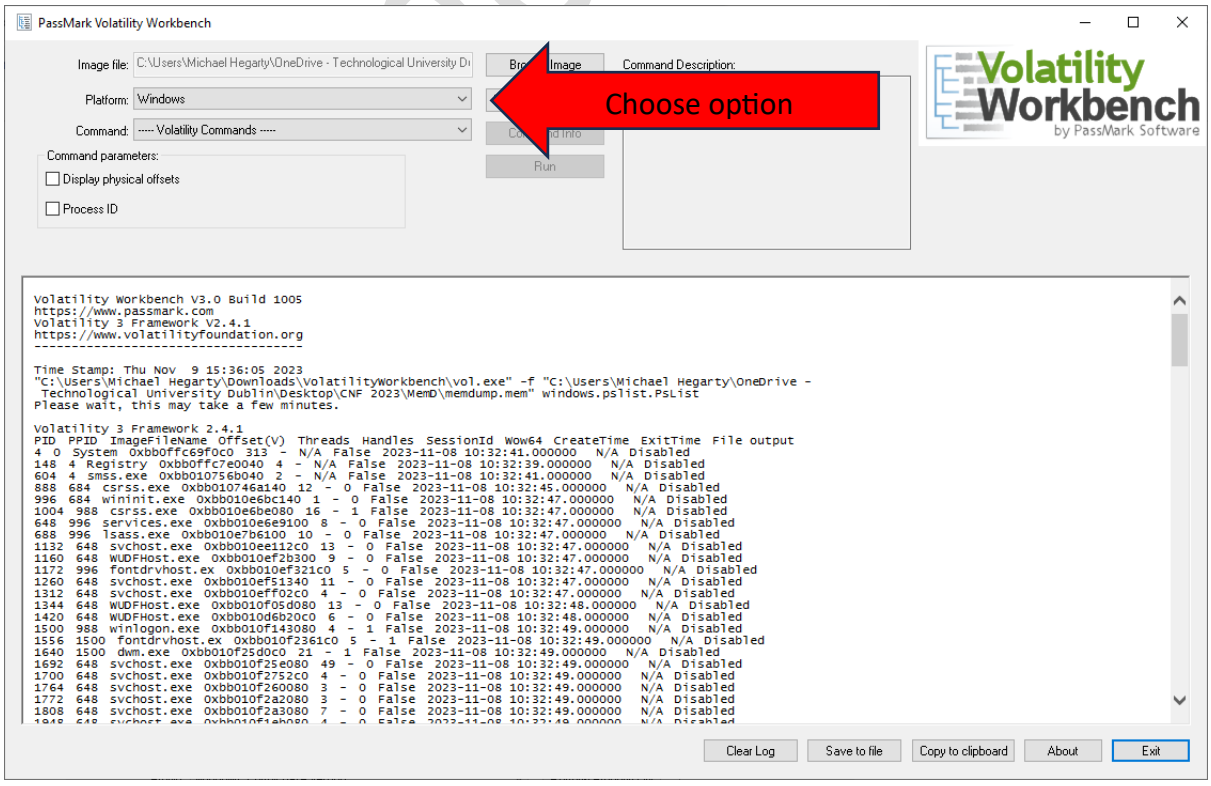
1. Eliminating the need to memorize complex command-line parameters.
2. The ability to store the operating system profile, KDBG address, and the process list within a .CFG file. This feature significantly streamlines the process of reloading a memory image, saving time and reducing the risk of profile selection errors.
3. Simplifying tasks such as copying, pasting, and printing, accessible via right-click options.
4. Streamlining the process of saving dumped information to a file on the disk.
5. Providing a dropdown list of available commands with concise descriptions of their functions.
6. Automatically timestamping the executed commands.
7. Offering the convenience of auto-loading the first memory dump file found in the current working directory.
8. Extending support for the analysis of Mac and Linux memory dumps, with profiles stored in the designated Profiles folder.

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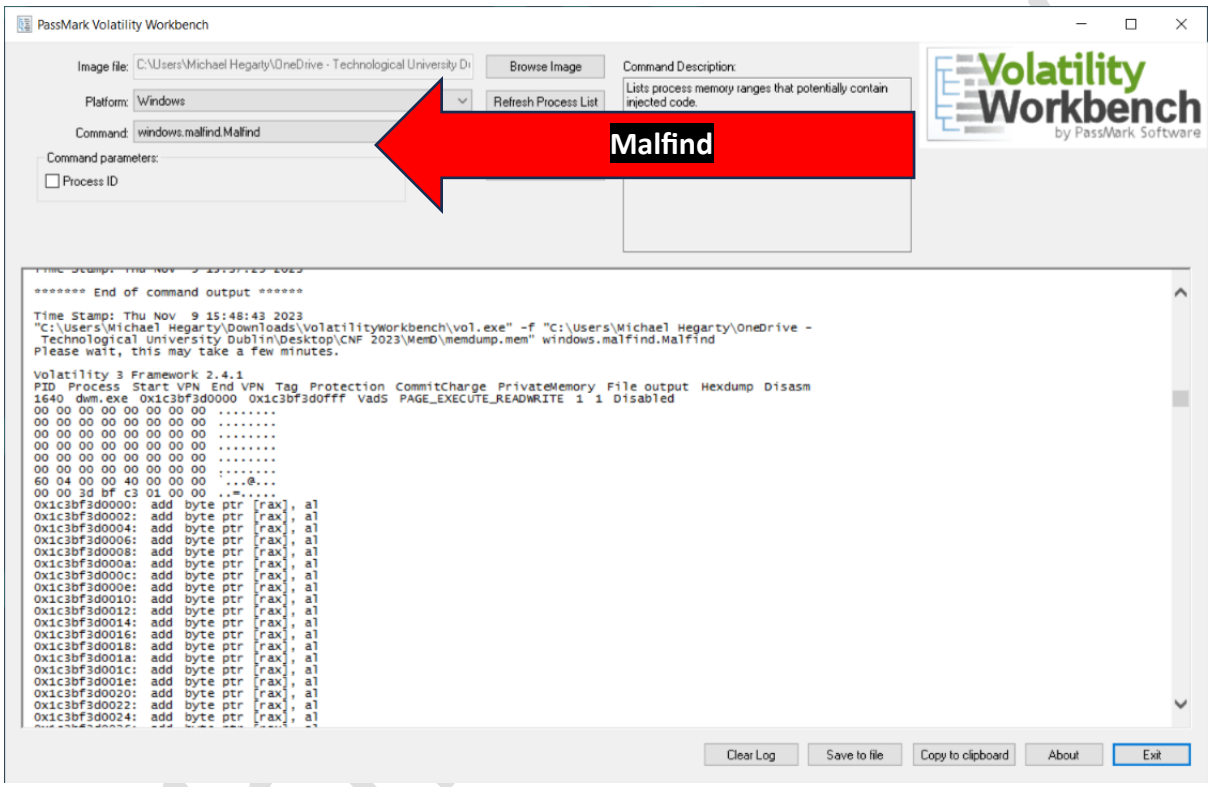
Upload the memory dump you created with FTK IMAGER





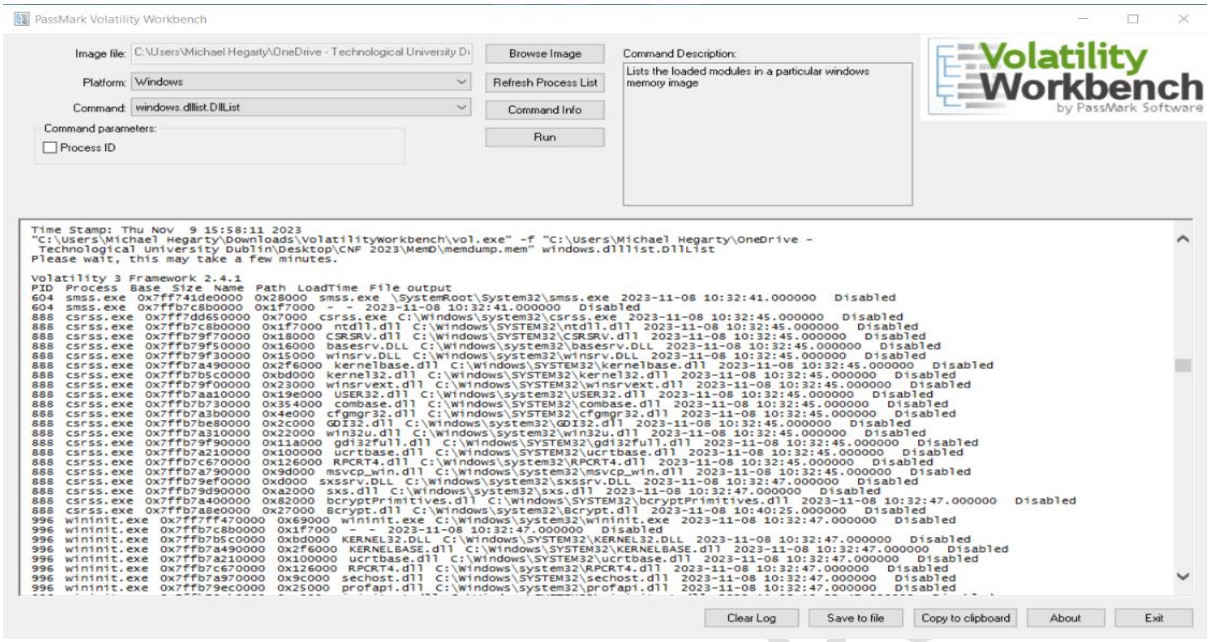
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Malfind is a specialized command used in memory analysis for the purpose of identifying concealed or injected code within a user's memory. Unlike its primary function of detecting the mere presence of a Dynamic Link Library (DLL) in a running process, Malfind focuses on pinpointing the exact location of these injected DLLs.



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DllList



These are just some example commands to run.

Please explore the tool further and learn by doing.

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